



# Bronchiectasis in Older Adults: Distinctive Epidemiology, Etiology, Diagnosis, and Clinical Evolution Compared with Younger Populations

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## Abstract

Non-cystic fibrosis bronchiectasis is increasingly recognized as a major cause of chronic respiratory morbidity, particularly among older adults. Its rising prevalence is largely attributable to improved diagnostic imaging and heightened clinical awareness, with epidemiological data consistently demonstrating a marked age-related increase in disease burden. Older patients represent the majority of contemporary cases and exhibit a distinct clinical phenotype characterized by multimorbidity, frailty, and worse outcomes. Pathophysiologically, bronchiectasis is driven by a 'vicious vortex' of airway inflammation, impaired mucociliary clearance, chronic infection, and structural lung damage. In older individuals, this process is amplified by immunosenescence, inflammaging, and age-related structural and functional lung changes, which together promote persistent infection and progressive airway injury. Additional contributors, such as microaspiration, cumulative environmental exposures, and coexisting conditions, including chronic obstructive pulmonary disease and gastroesophageal reflux disease, further exacerbate disease progression. Diagnosis in older adults with bronchiectasis is frequently delayed due to atypical presentations and clinical overlap with other chronic conditions, resulting in more advanced disease at the time of recognition. Microbiologically, older patients show a shift toward more virulent and resistant pathogens, particularly *Pseudomonas aeruginosa* and Enterobacteriaceae, with important implications for management. Prognosis is poorer in this population, with higher mortality, increased healthcare utilization, and reduced quality of life, largely driven by frailty and comorbidities rather than by disease severity alone. These findings highlight the need for a comprehensive, geriatric-informed approach integrating respiratory care with systematic assessment of functional status, comorbidities, and patient-centered outcomes to optimize management in older patients with bronchiectasis.

## 1 Introduction

Non-cystic fibrosis (CF) bronchiectasis is increasingly recognized as a major contributor to chronic respiratory morbidity worldwide [1]. Once considered a rare condition, its prevalence has risen substantially in recent decades [2], largely attributable to the widespread adoption of high-resolution computed tomography (HRCT) and heightened clinical awareness [3].

Historically, bronchiectasis was primarily associated with severe childhood respiratory infections or congenital disorders and was therefore considered a disease of younger populations. However, this paradigm has shifted. Epidemiological evidence now demonstrates that bronchiectasis is predominantly a disease of older adults, with prevalence increasing markedly with advancing age rather than being confined to younger cohorts [4]. Notably, prevalence rises from 7 per 100,000 individuals aged 18–34 years to 812 per 100,000 among those aged ≥75 years, underscoring the

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## Key Points

Bronchiectasis is predominantly a disease of older adults, with prevalence rising sharply with age due to increased diagnosis, longer survival, and accumulation of comorbidities such as chronic obstructive pulmonary disease and gastroesophageal reflux disease.

Aging-related biological changes drive disease severity, including immunosenescence (weakened immunity), inflammaging (chronic low-grade inflammation), and structural lung decline, which together worsen infection, mucus retention, and airway damage.

Older patients have worse outcomes due to delayed diagnosis and multimorbidity, with atypical symptoms, frequent misdiagnosis, higher risk of chronic infections (e.g., *Pseudomonas aeruginosa*), increased exacerbations, and higher mortality and healthcare burden.

Treatment in older adults with bronchiectasis must be individualized, balancing efficacy and safety due to frailty, comorbidities, and polypharmacy, with careful use of antibiotics and airway clearance strategies.

substantial healthcare burden associated with aging populations [1]. This demographic transition reflects both increased life expectancy and broader access to diagnostic imaging. However, it also highlights important age-related differences in disease susceptibility, clinical presentation, and outcomes, although the relative contribution of chronological age, biological aging, frailty, and multimorbidity remains incompletely understood.

The evolving demographic and clinical profile of bronchiectasis, therefore, raises important questions regarding age-related susceptibility, interactions with comorbid chronic diseases, and the implications for disease progression. This review aims to provide a comprehensive overview of bronchiectasis in older adults, with particular emphasis on pathophysiological mechanisms, clinical features, microbiology, and implications for its management.

## 2 Search Strategy and Evidence Synthesis

This article is a narrative review. A comprehensive literature search was conducted using major biomedical databases, including PubMed, ScienceDirect, Google Scholar, Mendeley, Scopus, Web of Science, and the Cochrane Library. The search covered publications up to March 2026. The search strategy combined controlled vocabulary terms and free-text keywords related to the disease and interventions of interest. These included, but were not limited to ‘bronchiectasis’, ‘non-cystic fibrosis bronchiectasis’, ‘elderly’, ‘older adults’, ‘aging’,

‘ageing’, ‘frailty’, ‘immunosenescence’, ‘inflammaging’, ‘multimorbidity’, ‘comorbidity’, ‘chronic infection’, ‘*Pseudomonas aeruginosa*’, ‘nontuberculous mycobacteria’, ‘microbiome’, ‘aspiration’, ‘gastroesophageal reflux disease’, ‘chronic obstructive pulmonary disease’, ‘COPD-bronchiectasis overlap’, ‘airway clearance’, ‘mucolytics’, ‘hypertonic saline’, ‘macrolides’, ‘inhaled antibiotics’, ‘polypharmacy’, ‘deprescribing’, ‘pulmonary rehabilitation’, ‘Comprehensive Geriatric Assessment’, ‘quality of life’, ‘hospitalization’, ‘mortality’, and ‘treatable traits’. Reference lists of relevant articles, reviews, guidelines, and registry reports were also manually screened to identify additional pertinent studies. Priority was given to recent publications, international guidelines, large observational cohorts, registry-based analyses, and studies specifically addressing older populations with bronchiectasis. Throughout this review, the term ‘older adults’ generally refers to individuals aged  $\geq 65$  years. However, because published bronchiectasis studies have used different age thresholds, the original age categories adopted by individual studies are reported where appropriate.

Given the narrative nature of this review, formal systematic review methodology and meta-analytic techniques were not applied. Instead, evidence was synthesized through a critical and interpretative approach aimed at highlighting clinically relevant findings, pathophysiological mechanisms, therapeutic implications, and current knowledge gaps. Study selection was guided by scientific quality, originality, relevance to the review objectives, and applicability to the care of older adults with bronchiectasis.

However, it should be acknowledged that studies specifically addressing older patients with bronchiectasis are limited. Consequently, some conclusions and management considerations are necessarily informed by evidence derived from broader populations of older adults with chronic respiratory diseases. This limitation has been considered when interpreting the available evidence. Furthermore, most of the available evidence comes from European and North American cohorts and registries. Therefore, the epidemiological patterns, microbiological characteristics, and management considerations discussed in this review may primarily reflect high-income healthcare settings. Caution is warranted when extrapolating these findings to regions with a higher prevalence of post-tuberculosis bronchiectasis or different healthcare infrastructures.

## 3 Epidemiology and Disease Burden in Older Adults

Epidemiological studies consistently demonstrate that the prevalence of bronchiectasis increases with age, with the highest rates observed in individuals over 70 years of age.

In the United States, bronchiectasis affects approximately 701 per 100,000 people overall, with approximately 500,000 adults affected [1]. However, this figure increases dramatically with age, ranging from 43 per 100,000 people in those aged 45–54 years to 373 per 100,000 people in those aged 65–74 years and reaching 812 per 100,000 people in the group aged  $\geq 75$  years [1]. In 2017, researchers in Germany noted the highest prevalence among those aged 70–79 years: 332.6 per 100,000 males and 313.1 per 100,000 females [6]. Similarly, the highest prevalence in Italy was observed in individuals over 75 years of age, with 466 cases per 100,000 people [7].

Comparable patterns have been reported globally. In Catalonia, Spain, both prevalence and incidence increased with age and were highest in men over 65 years of age, with a prevalence reaching 152.9 per 10,000 inhabitants (equivalent to 1529 per 100,000) [8]. Large registry data further reinforce the central role of age: the European Multicentre Bronchiectasis Audit and Research Collaboration (EMBARC), including 16,963 patients, reports a median age of 67 years, with a substantial proportion of patients aged  $\geq 65$  years and many exceeding 75 years [9]. In the United Kingdom, the median age at diagnosis between 2004 and 2013 was approximately 62 years [10]. In one large British cohort, 60% of patients received their diagnosis at or beyond 70 years of age [11]. These findings confirm that older adults represent the majority of bronchiectasis patients in current clinical practice.

The age at diagnosis reflects this trend, with a mean or median age of diagnosis that typically falls within the sixth or seventh decade of life across multiple countries and increases over time [1, 6–11]. Among newly diagnosed Medicare patients, the mean age is 76 years [12].

Frailty is increasingly recognized as a key determinant of outcomes in bronchiectasis and should be considered a central geriatric construct when evaluating these patients. The reported prevalence is high, potentially exceeding 40% in patients with comorbid nontuberculous mycobacterial (NTM) lung disease. This highlights the relevance of frailty in complex respiratory populations [13]. Frailty in bronchiectasis is closely related to malnutrition, low body mass index (BMI), and sarcopenia. These conditions are associated with reduced lung function, increased hospitalization rates, and higher mortality. These overlapping conditions underscore the need for integrated nutrition- and physical activity-based interventions aimed at mitigating functional decline.

Importantly, frailty is independently associated with adverse clinical outcomes. In patients with bronchiectasis, frailty increases the risk of 30-day hospital readmission by 36% [14] and has also been associated with higher mortality in older populations with bronchiectasis [15]. These findings

suggest that frailty is an active driver of prognosis, not merely a marker of disease burden.

The development of frailty in patients with bronchiectasis is multifactorial. Contributing factors include reduced handgrip strength, frequent exacerbations, a higher symptom burden (as indicated by scores on tests such as the COPD Assessment Test), and physical inactivity [16]. Of these factors, physical inactivity appears to play a particularly important role in exacerbation risk and healthcare utilization. Frailty is also closely associated with sarcopenia and malnutrition [17].

Older and frailer patients tend to experience more severe symptoms and greater impairment in physical function than younger patients [13, 18], reinforcing the concept that respiratory outcomes in older adults are strongly influenced by systemic rather than purely pulmonary factors. In older patients (particularly those aged  $\geq 80$  years), respiratory complications and comorbid conditions such as COPD and cardiovascular disease are common. These conditions contribute to a poorer quality of life, increased hospitalizations, and higher mortality rates. Evidence suggests that age-related differences in outcomes are driven more by frailty and comorbidity burden than by bronchiectasis severity or microbial load [15, 18]. Therefore, chronological age alone may not define a distinct bronchiectasis phenotype, and the contributions of biological aging and frailty should be considered when interpreting age-related differences.

Overall, frailty is more prevalent in patients with chronic respiratory diseases than in the general population. It represents a key vulnerability factor associated with hospitalization, functional decline, and mortality [19]. These findings underscore the importance of systematically assessing frailty in bronchiectasis patients and implementing targeted multidimensional interventions.

## 4 Pathophysiological Features in Older Adults

The disease is characterized by a ‘vicious vortex’ in which four interrelated pathophysiological components—airway inflammation, impaired mucociliary clearance, airway infection, and structural lung damage—contribute to one another simultaneously, driving progressive airway damage and functional decline if not adequately controlled [20, 21]. This conceptual model has evolved from the traditional ‘vicious cycle’ to better capture the complex, multidirectional interactions among these pathophysiological processes [1, 20, 21].

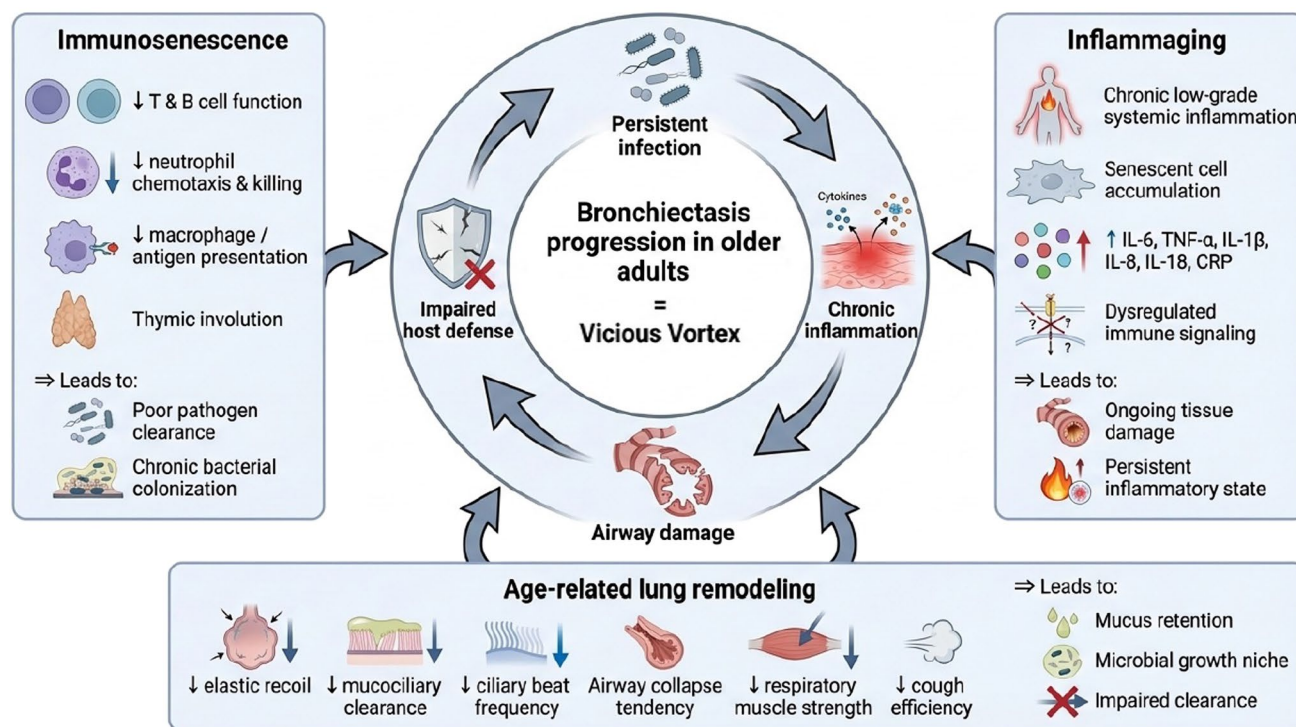
In older patients, three age-related biological processes amplify this ‘vicious vortex’ and contribute to more severe disease: (i) immunosenescence, the progressive decline

in immune function that impairs pathogen clearance and promotes chronic bacterial infection; (ii) inflammaging, the chronic low-grade systemic inflammation associated with aging that exacerbates airway damage; and (iii) structural and functional lung changes, including reduced elastic recoil, impaired mucociliary clearance, and diminished respiratory muscle strength, which together create a favorable environment for infection and mucus retention [22–24] (Fig. 1). In addition, cumulative lifetime exposures, such as recurrent respiratory infections, chronic airway diseases (particularly chronic obstructive pulmonary disease [COPD]), and environmental insults, further intensify these processes in older adults, reinforcing both susceptibility to infection and the persistence of inflammation [24]. These mechanisms do not act in isolation; rather, they interact synergistically to create a state of vulnerability in which the lungs of older adults are less able to mount effective host defenses, more prone to sustained and dysregulated inflammation, and structurally predisposed to progressive damage to the airways and worsening of the disease. Taken together, these observations suggest that biological aging influences how disease presents and progresses in older adults with bronchiectasis. However, it is unclear whether these differences represent a

true age-specific phenotype or the effects of frailty and multimorbidity.

#### 4.1 Immunosenescence

During adolescence, bronchiectasis is often linked to underlying conditions such as CF, primary ciliary dyskinesia, and immunodeficiencies, and is characterized by a robust inflammatory response involving neutrophils and elevated cytokine activity. If treated promptly, early-stage disease may show reversible features [25, 26]. In adulthood, the disease usually presents as an established condition [1], often idiopathic or secondary to infections such as tuberculosis or pneumonia. Inflammation remains predominantly neutrophilic, but is accompanied by more pronounced structural damage, with progression influenced by environmental and lifestyle factors, including smoking [1]. In older individuals, there is a shift toward altered innate and adaptive immune responses due to immunosenescence [27]. Immunosenescence encompasses multiple defects, including thymic involution, decreased T-cell and B-cell numbers and function, impaired phagocytosis and antigen presentation, and altered cellular metabolism [22]. Neutrophil dysfunction, including reduced chemotaxis and defective bacterial killing, has



**Fig. 1** Age-related mechanisms driving the vicious vortex of bronchiectasis progression in older patients. Schematic representation of how immunosenescence, inflammaging, and age-associated structural and functional lung changes interact with cumulative lifetime exposures

to amplify susceptibility to infection and sustain chronic inflammation in older adults with bronchiectasis. *CRP* C-reactive protein, *IL* interleukin, *TNF-α* tumor necrosis factor-α

been well documented [28]. Similarly, macrophage activity and antigen presentation decline with age, compromising host defense mechanisms [29]. In bronchiectasis, these alterations contribute to persistent bacterial infection and recurrent infections, which are key drivers of disease progression [30, 31].

## 4.2 Inflammaging

Aging is characterized by chronic low-grade systemic inflammation, known as inflammaging, which occurs in parallel with immunosenescence [32]. Inflammaging is driven by multiple factors, including accumulation of senescent cells, damage-associated molecular patterns from damaged cells and cell debris, and dysregulated immune responses [22]. Elevated circulating levels of pro-inflammatory cytokines, such as interleukin (IL)-6, tumor necrosis factor- $\alpha$ , IL-1 $\beta$ , IL-8, IL-13, IL-18, and C-reactive protein, have been implicated in tissue damage and disease progression [33]. The combination of immunosenescence (impaired immune defense) and inflammaging (persistent inflammation) exacerbates bronchiectasis pathology in older patients [27, 34]. This interaction contributes to a vicious vortex in which impaired pathogen clearance leads to persistent infection, driving chronic inflammation and further immune dysfunction [1]. Cellular senescence is a key mediator of lung aging and susceptibility to infection, with senescent cells accumulating in the respiratory system and contributing to both structural deterioration and immune dysregulation [23].

## 4.3 Structural and Functional Lung Changes

Age-related structural changes in the lungs further contribute to disease severity. These changes include reduced elastic recoil due to enlarged alveolar air spaces and redistribution of collagen and elastin, decreased mucociliary clearance resulting from lower ciliary beat frequency and impaired mucus transport, increased airway collapsibility secondary to loss of supporting tissue, and chest wall stiffening with decreased respiratory muscle strength [35–37]. Such alterations promote mucus retention and create a favorable environment for microbial infection, thereby perpetuating the inflammatory cycle characteristic of bronchiectasis [1, 2, 20]. Additionally, age-related decreases in cough effectiveness and respiratory muscle strength further impair airway clearance mechanisms, increasing the risk of recurrent infections [38]. The combination of impaired mucociliary clearance, structural lung changes, and increased susceptibility to aspiration in older adults creates a particularly vulnerable environment for bronchiectasis development and progression [24].

## 4.4 Aspiration-Related Bronchiectasis in Older Adults

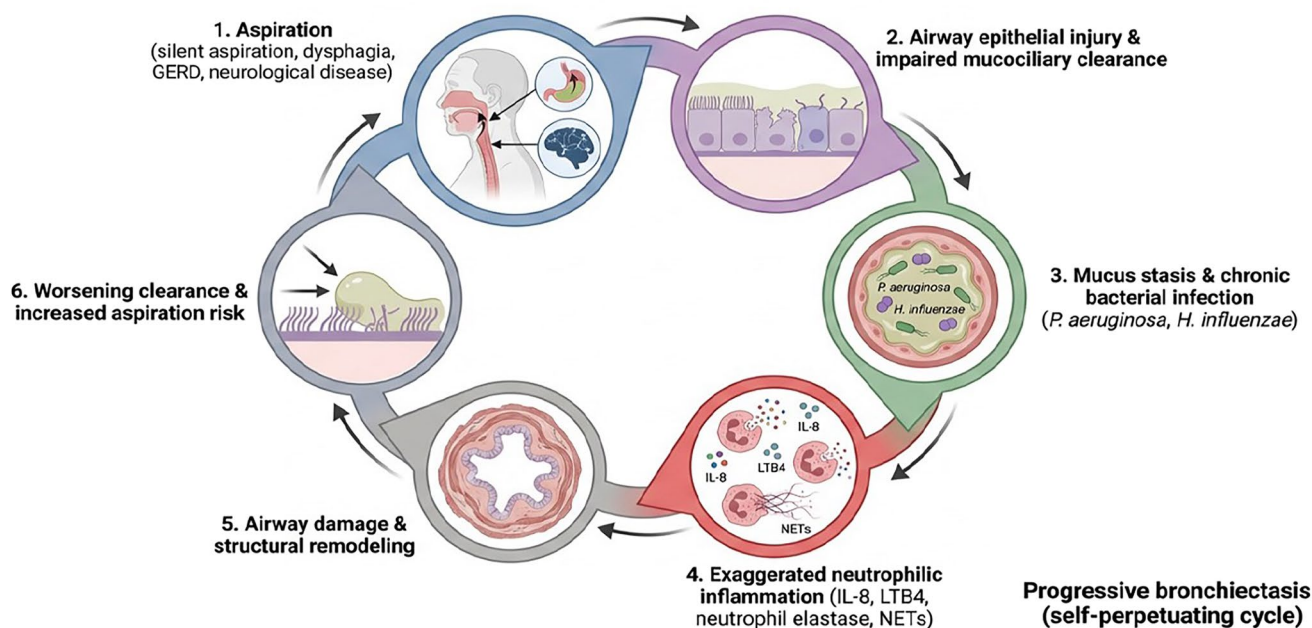
Aspiration-related bronchiectasis develops when recurrent or chronic microaspiration of oropharyngeal or gastric contents results in persistent airway injury, chronic infection, and progressive bronchial damage [20] (Fig. 2). Older adults are at increased risk due to the higher prevalence of dysphagia, neurological disorders such as stroke and Parkinson's disease, and gastroesophageal reflux disease (GERD) [39]. Age-related changes in swallowing function (presbyphagia), reduced cough reflex sensitivity, and impaired airway protective mechanisms further increase susceptibility to silent aspiration and chronic airway injury [24, 39].

Repeated microaspiration disrupts the bronchial epithelium and impairs mucociliary clearance. This promotes mucus retention and bacterial infection, particularly by pathogens such as *Pseudomonas aeruginosa* and *Haemophilus influenzae* [1, 20]. A predominantly neutrophilic inflammatory response is triggered by this process, and it is mediated by cytokines and chemotactic factors, including IL-8 and leukotriene B4 [1, 40]. Activated neutrophils release proteolytic enzymes and neutrophil extracellular traps (NETs), contributing to tissue damage, mucus hypersecretion, and persistent airway inflammation [2, 40, 41].

Over time, chronic infection and inflammation can lead to extracellular matrix degradation, airway wall injury, and structural remodeling. This process can result in irreversible bronchial dilatation, which is a hallmark of bronchiectasis [2, 16, 41, 42]. These mechanisms may be particularly relevant in older adults due to age-related declines in mucociliary function and cough effectiveness, which amplify the consequences of recurrent aspiration [24, 43].

## 4.5 Etiological Shift: Reduced Frequency of Genetic and Immunologic Causes in Older Adults

Genetic and immunologic causes of bronchiectasis, such as primary ciliary dyskinesia, congenital immunodeficiencies, and other inherited disorders, are far less common in older adults than in children or younger adults [1]. These conditions usually manifest early in life, often prompting diagnosis and management during childhood or young adulthood, and many affected individuals either do not survive to advanced age or are already identified and treated before reaching older age. In children, these etiologies account for up to 13–19% of bronchiectasis cases [40]. In older populations, bronchiectasis more often reflects the cumulative effects of environmental exposures and prior infections, rather than congenital or



**Fig. 2** The aspiration–infection–inflammation vicious cycle in older adults with bronchiectasis. Recurrent (often silent) microaspiration leads to epithelial injury, impaired mucociliary clearance, infection, and neutrophilic inflammation, driving progressive airway damage

and structural remodeling in a self-perpetuating cycle. *GERD* gastroesophageal reflux disease, *IL* interleukin, *LTB4* leucotriene B4, *NETs* neutrophil extracellular traps

inherited disorders [2]. Moreover, the natural course of many congenital conditions tends to cause more severe disease and earlier morbidity, further reducing their prevalence in older patients [42]. This etiological shift with age is consistently observed in multicenter registries and clinical reviews.

#### 4.6 Comorbidities

Comorbidities are common in bronchiectasis and are associated with higher mortality and poorer quality of life [21]. Conditions such as COPD, GERD, and cardiovascular disease are more frequent in older patients, contributing to disease progression and clinical complexity, and are less pronounced in younger populations.

The cumulative impact of COPD-related airway damage over time, driven by chronic airway inflammation, repeated exacerbations, and progressive structural remodeling, is a major contributor to bronchiectasis in older adults. The overlap between COPD and bronchiectasis is especially common in older individuals [44], with a prevalence ranging from 20% to 69% depending on the population studied [45, 46]. According to US registry data, approximately 20% of patients with non-CF bronchiectasis have concurrent COPD [1]. The wide variation in reported prevalence likely reflects methodological differences across studies, including variations in study populations, disease severity,

healthcare settings, and diagnostic definitions. Some studies define overlap based primarily on radiological evidence of bronchiectasis in patients with COPD, while others require both radiological bronchiectasis and spirometrically confirmed airflow obstruction. Furthermore, the label of COPD is often applied to patients with bronchiectasis who lack objective evidence of airflow obstruction or a smoking history, which influences prevalence rates [46]. Consequently, prevalence estimates should be interpreted with caution, as differences in case definitions can substantially influence reported rates.

Despite this variability, COPD–bronchiectasis overlap represents a clinically important phenotype. It is associated with an increased risk of exacerbations (odds ratio [OR] 1.97), isolation of pathogenic microorganisms (OR 4.11), severe airflow obstruction (OR 1.31), and mortality (OR 1.96) [47]. Patients with COPD–bronchiectasis overlap also experience more hospitalizations and incur higher healthcare costs than those with COPD alone [48]. These findings suggest that the coexistence of COPD and bronchiectasis identifies a subgroup of patients with a greater disease burden and poorer outcomes. Furthermore, because both disorders commonly affect older individuals and share several clinical manifestations, their coexistence may complicate diagnosis and contribute to delays in appropriate disease recognition.

GERD is the most common comorbidity (up to 47% of patients in US registries [1]), with prevalence increasing with age [49]. It contributes via aspiration and vagally mediated airway spasm [50]. GERD is associated with higher mortality, greater severity, more exacerbations and hospitalizations, worse lung function, and reduced quality of life [51]. GERD also increases the risk of developing bronchiectasis (hazard ratio [HR] 1.43 overall; up to 8.46 in those aged  $\geq 70$  years) [46] and raises healthcare utilization and costs [52].

Bronchiectasis patients have increased cardiovascular risk, with higher prevalence of coronary heart disease (OR 1.33) and stroke (OR 1.92), and increased incident events (HR 1.44 and 1.71, respectively) [53]. Events during/after exacerbations raise long-term mortality [54]. Risk factors include age, hypertension, COPD, and severe exacerbations [54].

Common additional conditions include osteoporosis, depression, anxiety, chronic rhinosinusitis, malnutrition, and autoimmune diseases such as rheumatoid arthritis [21]. Guidelines recommend systematic assessment of comorbidities in all bronchiectasis patients, including screening for cardiovascular disease, osteoporosis, anxiety/depression, rhinosinusitis, and GERD [21].

## 5 Clinical Presentation and Diagnostic Challenges

Chronic productive cough and recurrent exacerbations are hallmark features of bronchiectasis in all age groups [1, 2]. However, the clinical presentation varies by age due to differences in symptom reporting, comorbidity burden, and disease etiology. Diagnosis in older patients is often substantially delayed due to atypical presentations, attribution of symptoms to aging or comorbidities, and clinical overlap with other common respiratory conditions (Table 1). This results in a more advanced disease at the time of diagnosis [1, 2, 24].

### 5.1 Core Symptoms Across All Ages

The universal features include chronic cough (lasting  $>8$ –12 weeks) with daily sputum production, recurrent respiratory exacerbations, dyspnea, and fatigue [1, 2]. Sputum characteristics have prognostic significance: purulent sputum (versus mucoid) is associated with higher exacerbation rates (incident rate ratio 1.55), more hospitalizations (incident rate ratio 1.98), and increased mortality (HR 1.12) [1]. Physical examination may reveal prolonged exhalation with adventitious sounds including crackles (40%), rhonchi (22%), and wheezes (16%) [1]. However, some patients present with

radiographic bronchiectasis as an incidental finding with minimal or no symptoms [2].

### 5.2 Atypical Presentation in Older Patients

The prevalence of classic respiratory symptoms in older patients does not differ significantly from that in younger adults [15]. Cough, sputum production, and hemoptysis occur at comparable rates across adult age groups. However, the clinical presentation in this population is often atypical, characterized by non-specific manifestations that can complicate timely diagnosis [2, 24].

Specifically, older individuals often present with fatigue and reduced functional capacity, which are commonly attributed to deconditioning [2]. Dyspnea is also frequently reported, yet it is often misinterpreted as a symptom of comorbid conditions, such as heart failure or COPD [2]. Similarly, weight loss may be erroneously attributed to malignancy or the aging process itself [2]. These features may be accompanied by systemic symptoms, including intermittent fever, night sweats, chest pain, and shortness of breath [2].

The nonspecific nature of these symptoms often leads to misattribution to normal aging or preexisting comorbidities, such as COPD, heart failure, or chronic bronchitis, resulting in substantial diagnostic delays [2, 24]. Furthermore, older patients may underreport symptoms due to reduced physical activity levels or the perception that respiratory complaints are an expected consequence of aging [24].

### 5.3 Magnitude of Diagnostic Delay

Diagnostic delays are a pervasive issue across all age groups, particularly affecting older adults. A survey of 760 patients from 40 countries revealed that 15% had experienced symptoms for 5–10 years before receiving an accurate diagnosis, and an additional 25% had symptoms for over ten years before diagnosis [1].

Earlier diagnosis is more common in younger populations, in whom a higher index of clinical suspicion and more specific symptoms facilitate timely recognition [56]. Conversely, older patients more often present with advanced disease, which is frequently idiopathic or secondary to previous infectious processes [56].

### 5.4 Clinical Overlap with Other Respiratory Conditions

Significant clinical overlap exists between bronchiectasis and other common respiratory conditions in older patients, particularly COPD, asthma, and chronic rhinosinusitis, which substantially complicates diagnosis [2, 24]. Diagnosis

**Table 1** Clinical presentation and diagnostic challenges of bronchiectasis in older adults

| Domain                                | Key features                                                                         | Clinical implications in older adults                                                                                |
|---------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Core symptoms (all ages)              | Chronic cough (>8–12 weeks), daily sputum, dyspnea, fatigue, recurrent exacerbations | Universal presentation; purulent sputum associated with worse outcomes (↑ exacerbations, hospitalization, mortality) |
| Physical findings                     | Crackles, rhonchi, wheeze; sometimes minimal signs                                   | Often non-specific or mild despite significant disease                                                               |
| Atypical presentation in older adults | Fatigue, reduced function, weight loss, dyspnea, systemic symptoms                   | Frequently misattributed to aging or comorbidities (COPD, HF, malignancy) → diagnostic confusion                     |
| Symptom perception/reporting          | Underreporting of respiratory symptoms                                               | Reduced clinical suspicion and delayed referral for imaging                                                          |
| Diagnostic delay                      | Symptoms often present for years before diagnosis                                    | Long delay (up to >10 years in 25% of patients) → more advanced disease at diagnosis                                 |
| Clinical overlap                      | COPD, asthma, chronic bronchitis, chronic rhinosinusitis                             | Frequent misdiagnosis; empirical treatments (antibiotics, ICS, bronchodilators) before correct diagnosis             |
| Radiological findings                 | Bronchiectasis often incidental on CT                                                | Diagnosis requires correlation of symptoms + imaging; incidental findings may be overlooked                          |
| Consequences of delay                 | Greater structural lung damage, reduced lung function                                | Higher disease burden, worse functional status, increased exacerbations                                              |
| Chronic infection risk                | Increased likelihood of <i>Pseudomonas aeruginosa</i> infection                      | Marker of severe disease; associated with faster decline and higher mortality                                        |
| Prognosis impact                      | Frequent exacerbations and reduced FEV <sub>1</sub>                                  | Increased mortality risk and poor long-term outcomes                                                                 |

*COPD* chronic obstructive pulmonary disease, *CT* computed tomography, *FEV<sub>1</sub>* forced expiratory volume in 1 second, *HF* heart failure, *ICS* inhaled corticosteroids

is often delayed because older adults are incorrectly thought to have chronic bronchitis, chronic rhinosinusitis, or other causes of chronic cough [2]. The condition may be misdiagnosed as COPD or asthma until appropriate testing has been performed [2]. In such cases, patients often receive multiple courses of empirical antibiotics, inhaled glucocorticoids, and bronchodilators before the correct diagnosis is considered and confirmed [2]. Radiographic bronchiectasis is a common incidental finding in older adults undergoing CT imaging for other indications; however, clinical diagnosis of bronchiectasis requires both characteristic symptoms and radiographic findings [24]. When symptoms are attributed to known comorbidities rather than prompting a dedicated imaging evaluation, diagnosis may be further delayed.

### 5.5 Consequences of Delayed Diagnosis in Older Patients

Delayed recognition of bronchiectasis in older adults is associated with more advanced disease at diagnosis and worse clinical outcomes [1, 2, 55]. Prolonged symptom duration correlates with greater radiological extent, impaired lung function, and increased rates of bacterial infection, as shown in the EMBARC registry [55].

Late diagnosis also facilitates chronic bacterial infection, particularly with *P. aeruginosa*, a key marker of disease

severity linked to more frequent exacerbations, accelerated lung function decline, and increased mortality [1, 2].

Exacerbations themselves contribute to progressive functional deterioration and reduced quality of life; patients with frequent exacerbations (≥3 per year) have the highest risk of recurrence and 5-year mortality [1, 2]. Declining forced expiratory volume in one second (FEV<sub>1</sub>) is an independent predictor of mortality in older patients [15].

Overall, delayed diagnosis is associated with poorer prognosis, especially in the presence of recurrent exacerbations, chronic infection, and comorbidities such as COPD, highlighting the importance of earlier recognition and heightened clinical awareness [1].

### 5.6 Limitations of Current Severity Scores and the Role of Geriatric Assessment

Several multidimensional indices are widely used to assess disease severity and prognosis in bronchiectasis (Table 2). These include the Bronchiectasis Severity Index (BSI), which incorporates age as a direct scoring component alongside BMI, FEV<sub>1</sub>, previous hospitalizations, exacerbation frequency, radiological extent of disease, and chronic infection with *P. aeruginosa* or other pathogens; the FACED score, based on FEV<sub>1</sub>, age, chronic *P. aeruginosa* infection, radiological extension, and dyspnea; and the E-FACED score,

which additionally includes exacerbation history [57, 58]. While these tools integrate key clinical, microbiological, and physiological variables, they all include age as a weighted component, which may introduce confounding factors in older populations. Consequently, older patients may be misclassified as having more severe disease due to their age rather than their bronchiectasis-specific disease burden.

This limitation is particularly relevant in the context of the aforementioned diagnostic delay, wherein older adults often present with more advanced disease and a higher burden of multimorbidity. Consequently, distinguishing between disease severity, accumulated comorbidity, and age-related vulnerability remains challenging.

In this context, frailty has emerged as a complementary prognostic construct reflecting biological rather than chronological aging. The Comprehensive Geriatric Assessment (CGA) is a structured, multidimensional process that evaluates medical, functional, psychological, and social domains. It is commonly used to identify frailty-related deficits and care needs in older adults [59]. The CGA helps to identify potentially modifiable factors contributing to functional decline and supports the development of individualized care plans. Furthermore, frailty indices derived from the CGA, such as the Multidimensional Prognostic Index (MPI), have demonstrated strong predictive accuracy for mortality and adverse outcomes, including hospitalization and institutionalization, in older populations [60].

In addition to prognostic indices and frailty assessment, patient-reported outcome measures (PROMs) provide valuable insights into the impact of bronchiectasis on daily functioning and health-related quality of life [61]. Validated bronchiectasis-specific instruments, such as the Quality of Life–Bronchiectasis (QoL-B) questionnaire and the Bronchiectasis Health Questionnaire (BHQ), capture dimensions that are not adequately reflected by physiological or severity-based scores [61]. Incorporating these instruments into clinical assessments may support a more comprehensive and patient-centered evaluation of older adults with bronchiectasis.

Therefore, integrating frailty assessment and CGA and established severity indices with PROMs may provide a more clinically meaningful approach to risk stratification in older adults with bronchiectasis. This approach supports a shift toward a more comprehensive, patient-centered model of care.

# 6 Microbiology in Bronchiectasis: Special Considerations in Older Adults

Chronic bacterial infection is a central feature of bronchiectasis pathophysiology, and microbiological assessment is essential for guiding treatment and determining prognosis

**Table 2** Multidimensional bronchiectasis severity indices and key limitations in older adults

| Score         | Components                                                                                                                                           | Primary purpose                                              | Strengths                                                                   | Limitations in older adults                                                                                         |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| BSI           | Age, BMI, FEV <sub>1</sub> , hospitalizations, exacerbation frequency, radiological extent, chronic <i>Pseudomonas aeruginosa</i> or other pathogens | Predicts mortality, exacerbations, hospitalization risk      | Comprehensive, well validated, good prognostic accuracy                     | Age is directly weighted, which may inflate severity classification in older patients independent of disease burden |
| FACED score   | FEV <sub>1</sub> , age, chronic <i>P. aeruginosa</i> infection, radiological extension, dyspnea                                                      | Classifies disease severity (mild/moderate/severe)           | Simple, easy to apply in clinical practice                                  | Age inclusion may confound severity assessment in older populations; less sensitive to exacerbation burden          |
| E-FACED score | FACED components + exacerbation history                                                                                                              | Improves prediction of exacerbation risk and hospitalization | Better prognostic performance than FACED; includes dynamic clinical history | Still includes age; residual confounding by chronological ageing remains                                            |

BMI body mass index, BSI Bronchiectasis Severity Index, FEV<sub>1</sub> forced expiratory volume in 1 second

[2, 20]. The term ‘chronic infection’ is preferred over ‘colonization’ because it more accurately reflects the long-term interaction between microorganisms and the host that leads to progressive tissue damage [20]. In older patients, age-related changes in immune function and airway defenses lead to a distinct microbiological profile, likely driven by immunosenescence, longer disease duration, repeated antibiotic exposure, and an increased risk of aspiration.

### 6.1 Age-Related Microbiological Profile in Bronchiectasis

Several studies suggest that there are age-associated shifts in microbiological patterns compared with those of younger adults, with a greater prevalence of more virulent and therapeutically challenging pathogens. These differences may have significant implications for disease severity, management of antimicrobials, and overall prognosis. However, they are likely influenced by factors such as frailty, burden of comorbidities, cumulative antibiotic exposure, and disease duration.

In patients aged  $\leq 64$  years, *H. influenzae* is the predominant organism [62] and is more commonly associated with acute exacerbations rather than chronic infection [63]. In contrast, among those aged  $>64$  years, *P. aeruginosa*, Enterobacteriaceae, and NTM become more prevalent [62].

Specifically, the prevalence of *P. aeruginosa* increases significantly with age (OR 1.027 per year;  $p = 0.008$ ) [1]. Age  $>55$  years is an independent predictor of chronic *P. aeruginosa* infection [64]. Chronic *P. aeruginosa* infection is associated with more severe disease overall, including more frequent exacerbations, a higher risk of hospitalization, a worse quality of life, increased mortality, more extensive radiographic involvement, and an accelerated decline in lung function [63].

Enterobacteriaceae (including *Klebsiella* species and *Escherichia coli*) are significantly more prevalent in older patients (OR 1.039 per year;  $p = 0.01$ ) [62]. The increasing prevalence of Enterobacteriaceae in older adults reflects multiple converging factors, including increased aspiration risk due to dysphagia and GERD [65], enhanced gut–lung axis communication with aging [66], and cumulative antibiotic exposure that selects for resistant Gram-negative organisms [67].

NTM deserve special consideration in older patients with bronchiectasis because advancing age is a recognized risk factor for NTM pulmonary disease [62]. The most frequently isolated organisms are members of the *Mycobacterium avium* complex. However, the *Mycobacterium abscessus* complex is less common yet more challenging to treat due to its intrinsic antimicrobial resistance [68]. The increased susceptibility in older adults likely reflects immunosenescence, impaired mucociliary clearance, structural airway

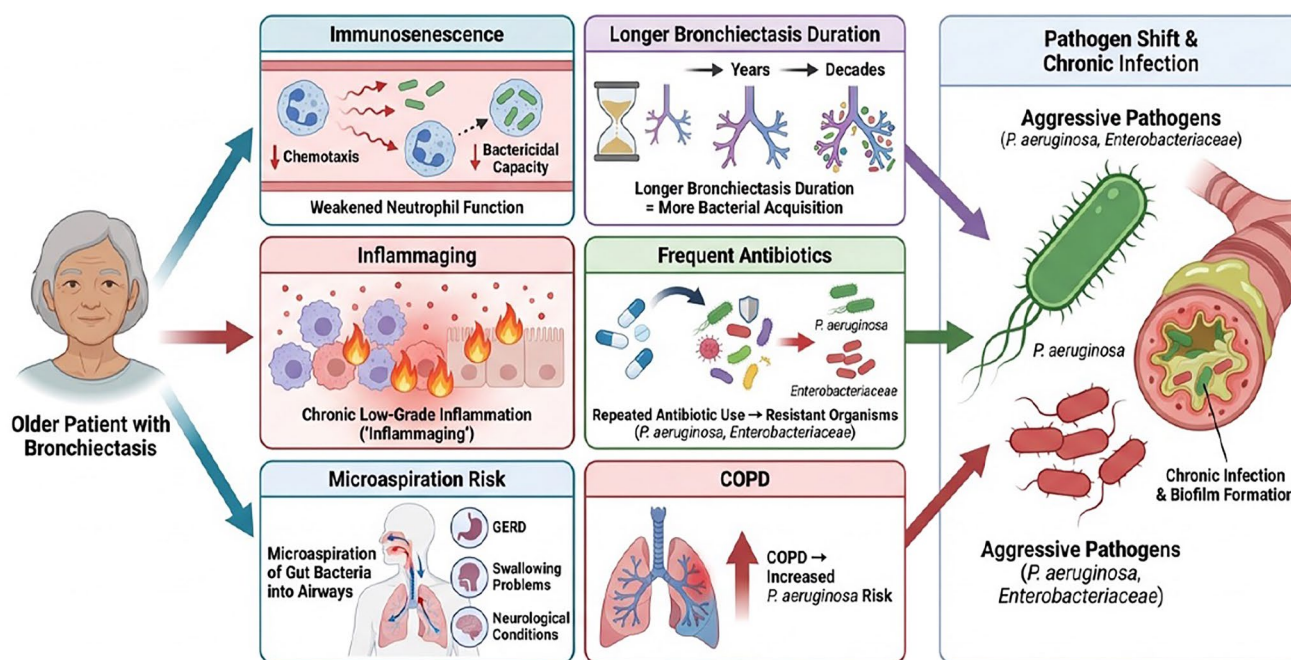
abnormalities, and cumulative environmental exposure [69]. NTM pulmonary disease typically presents with a nodular-bronchiectatic pattern on HRCT, characterized by bronchiectasis involving the right middle lobe and lingula, associated nodules, and tree-in-bud opacities [68]. However, diagnosis may be challenging because symptoms and radiological findings frequently overlap with those of bronchiectasis itself [70]. Importantly, NTM isolation does not necessarily indicate active disease, and diagnosis requires the integration of clinical, radiological, and microbiological criteria [69]. The management of NTM infection is particularly challenging in older adults because treatment is prolonged and associated with significant toxicity, including hepatotoxicity, ototoxicity, visual impairment, and drug–drug interactions related to polypharmacy [69, 71]. Due to its potential impact on disease progression and therapeutic decisions, mycobacterial cultures should be considered for patients with suggestive clinical or radiological features and repeated when clinically indicated.

### 6.2 Mechanisms Underlying Age-Related Pathogen Shift

In older patients, the shift toward more aggressive pathogens appears to result from several age-related changes occurring together (Fig. 3). The immune system becomes less effective over time, with neutrophils showing impaired chemotaxis and reduced bactericidal capacity, which makes chronic infections with *P. aeruginosa* more likely to become established [28]. At the same time, there is a state of chronic low-grade inflammation, known as inflammaging, characterized by persistently elevated inflammation despite reduced capacity to combat infections [72]. Older patients also tend to have had bronchiectasis for a longer duration, providing more opportunities for virulent bacteria to be acquired and become persistent [65]. In addition, repeated antibiotic treatments over the years can select for more resistant organisms such as *P. aeruginosa* and Enterobacteriaceae [73], while age-related swallowing problems, GERD, and neurological conditions increase the risk of microaspiration, introducing gut bacteria into the lower airways [65, 74]. Finally, the higher prevalence of COPD in this population further increases the risk of *P. aeruginosa* infection [75].

### 6.3 Clinical Implications

The distinct microbiological profile seen in older patients with bronchiectasis has several practical consequences. During exacerbations, empiric antibiotic therapy should consider the higher likelihood of *P. aeruginosa* and Enterobacteriaceae, often justifying antipseudomonal coverage



**Fig. 3** Mechanisms underlying the age-related shift in airway pathogens in older patients. Interacting age-associated biological and clinical factors drive the emergence of more virulent and drug-resistant

pathogens in the airways of older individuals with bronchiectasis. *COPD* chronic obstructive pulmonary disease, *GERD* gastroesophageal reflux disease

when clinically appropriate. Regular sputum surveillance becomes especially important in this population to identify new acquisition of *P. aeruginosa*, which may warrant eradication strategies [21]. In patients with chronic *P. aeruginosa* infection and frequent exacerbations, long-term suppressive treatment with inhaled antibiotics or macrolides can be considered, although careful monitoring is needed due to potential adverse effects and drug interactions [21]. Finally, given the increased prevalence of NTM in older individuals, mycobacterial cultures should be performed at diagnosis and repeated over time, particularly when clinical or radiological features raise suspicion [21].

## 7 Practical Management Considerations in Older Adults with Bronchiectasis

The 2025 European Respiratory Society guidelines for adult bronchiectasis offer a comprehensive overview of disease management. This includes patient assessment, pharmacological and non-pharmacological treatments, and strategies for managing exacerbations and clinical deterioration [21]. While these recommendations apply to all age groups, they do not specifically address the unique clinical challenges faced by older adults with bronchiectasis (Table 3). To facilitate the application of these principles in geriatric care, Figure 4 presents a practical management pathway for older adults with bronchiectasis.

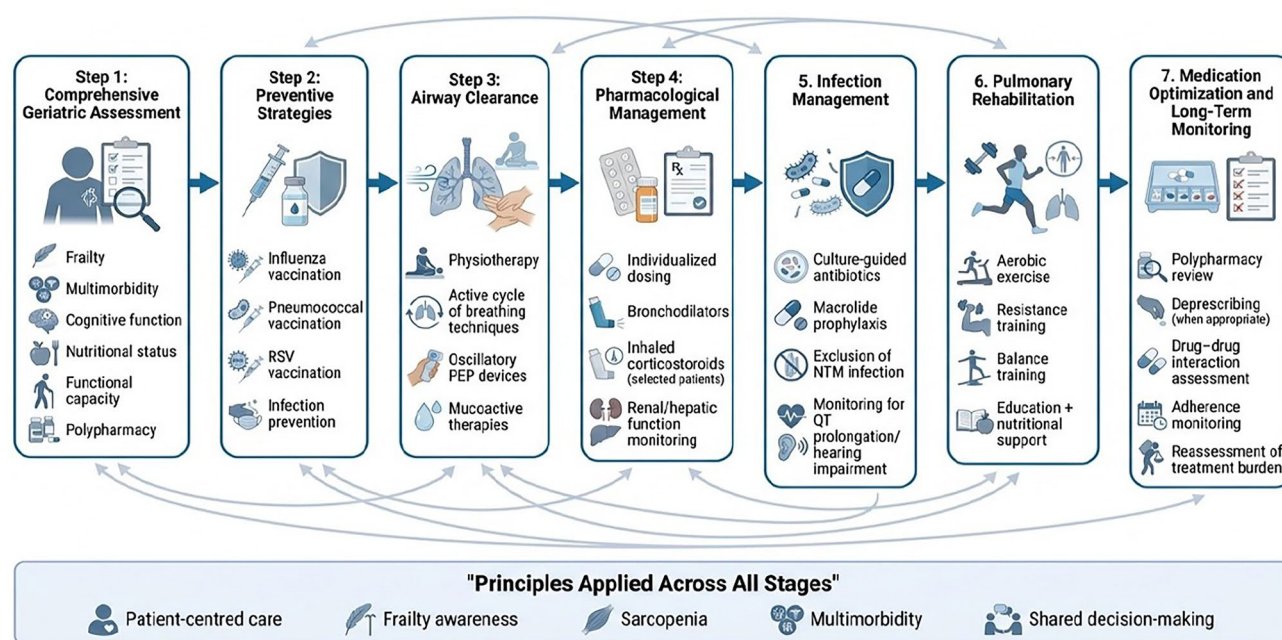
Preventive measures are an important part of managing older adults. Annual influenza vaccination and age-appropriate pneumococcal vaccination are recommended to reduce the risk of respiratory infections and their complications [76]. Respiratory syncytial virus (RSV) is a significant cause of severe respiratory illness in older adults. New vaccines offer a preventive strategy, particularly for those who are more vulnerable to severe lower respiratory tract infection [77]. Subunit vaccines, including those involving the pre-F conformation of the RSV fusion protein, have demonstrated significant protection in older adults, reducing the rate of total and severe RSV-related lower respiratory tract diseases [78]. Although evidence specific to bronchiectasis remains limited, with most research focusing on other chronic lung disorders [79], optimizing vaccination status is a low-risk intervention that may reduce respiratory morbidity and healthcare utilization.

The pharmacological management of older adults with bronchiectasis is complicated by the combined effects of age-related physiological changes, multimorbidity, frailty, and polypharmacy. These factors collectively increase treatment complexity and susceptibility to adverse drug reactions [24]. Aging is associated with reduced renal and hepatic clearance, altered body composition (increased fat mass, reduced lean body mass, and total body water), and prolonged drug half-lives, especially for lipophilic agents [80, 81]. Pharmacodynamic changes, including increased sensitivity to anticoagulants, psychotropic agents, and

**Table 3** Key pharmacological challenges in older adults with bronchiectasis

| Domain                       | Main issue in older adults                                       | Clinical impact in bronchiectasis                                       |
|------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------|
| Pharmacokinetics             | Reduced renal/hepatic clearance; altered body composition        | Higher drug exposure and toxicity risk → need for dose adjustment       |
| Pharmacodynamics             | Increased drug sensitivity (e.g., anticoagulants, psychotropics) | Greater risk of adverse drug reactions                                  |
| Immune ageing                | Immunosenescence and inflammaging                                | Altered treatment response and increased infection susceptibility       |
| Polypharmacy                 | High medication burden and drug–drug interactions                | Increased adverse events → structured review and deprescribing required |
| Airway clearance therapy     | Frailty, cognitive/physical impairment                           | Reduced adherence → need for simplified, individualized approaches      |
| Inhaled corticosteroids      | Frequent multimorbidity; infection susceptibility                | Restricted use; ↑ risk of pneumonia and NTM infection                   |
| Antibiotics (macrolides)     | Cardiac comorbidity and QT-prolonging co-medication              | Higher arrhythmic risk → ECG and safety monitoring essential            |
| Inhaled antibiotics          | Reduced inspiratory flow in frailty                              | Potential reduced lung deposition and efficacy                          |
| New anti-inflammatory agents | Limited geriatric evidence                                       | Uncertain long-term safety in older adults                              |
| Deprescribing                | Complex multimorbidity                                           | Essential but under-studied in bronchiectasis                           |

ECG echocardiogram, NTM nontuberculous mycobacterial



**Fig. 4** Practical management pathway for older adults with bronchiectasis. Management begins with a comprehensive geriatric assessment, which informs subsequent preventive, pharmacological, rehabilitative, and monitoring strategies. Although presented as a sequential pathway, all domains are closely interconnected and should

be revisited over time according to changes in clinical status, treatment response, frailty, multimorbidity, and patient preferences. NTM nontuberculous mycobacterial, PEP positive expiratory pressure, RSV respiratory syncytial virus

glucose-lowering therapies, further amplify safety concerns [80, 81]. These effects may be exacerbated by immunosenescence and inflammaging, which can alter treatment responses and increase vulnerability to treatment-related harm [24, 82]. Therefore, medication selection and dosing

must be tailored to each individual, with regular renal and hepatic function reviews and careful consideration of drug–drug interactions.

Airway clearance is a fundamental aspect of managing bronchiectasis and should be tailored to the individual's

symptom burden, physical function, cognitive status, and caregiver support [21, 83]. Physiotherapy techniques, such as the active cycle of breathing and oscillatory positive expiratory pressure devices, can be used alongside mucoactive therapies for patients with persistent sputum production or recurrent exacerbations [1, 2]. Hypertonic saline may improve mucus hydration and facilitate expectoration; however, cough, bronchospasm, and the burden of treatment can limit its tolerability in frail individuals [84]. Oral mucoactive agents such as N-acetylcysteine [85] and carbocisteine [84] are also commonly used in clinical practice. However, evidence supporting their clinical benefit remains limited. In a recent randomized trial involving patients with non-CF bronchiectasis, daily sputum production, and frequent exacerbations, neither hypertonic saline nor carbocisteine significantly reduced the rate of pulmonary exacerbations over 52 weeks [84]. Nevertheless, mucoactive agents may be considered on an individual basis, particularly when excessive sputum production contributes to symptoms or an impaired quality of life [21]. Treatment selection should balance potential symptomatic benefits against adherence challenges, tolerability, and overall treatment burden.

Bronchodilators are primarily recommended for patients with coexisting COPD or asthma [86]. Inhaled corticosteroids are generally reserved for patients with specific conditions, such as asthma overlap syndrome or eosinophilic inflammation [87]. This is because they are only moderately effective in treating bronchiectasis and can increase the risk of pneumonia and NTM infections in older adults [87]. Practical use of inhaled therapies may be challenging for frail patients due to visual impairment, reduced manual dexterity, cognitive dysfunction, and age-related reductions in inspiratory flow, which can compromise correct device handling and medication delivery [83]. Therefore, periodic assessment of inhaler technique and caregiver involvement may be necessary to optimize treatment effectiveness.

Exacerbations are usually treated with 14-day antibiotic regimens based on previous microbiological results and local resistance patterns [1, 2]. A systematic review and meta-analysis revealed that long-term prophylactic macrolide therapy reduces the rate of acute exacerbations in patients with bronchiectasis, including those with *P. aeruginosa* infection, for which current guidelines do not always recommend macrolides [88]. However, age-specific safety considerations are particularly relevant. Older adults may be at an increased risk for QT interval prolongation due to concomitant use of other QT-prolonging medications and a high prevalence of underlying cardiovascular disease [89]. Macrolides may also exacerbate pre-existing sensorineural hearing impairment; therefore, audiological monitoring should be considered when clinically appropriate [90]. Additionally, clinically relevant drug–drug interactions

may occur through cytochrome P450 inhibition with certain macrolides, such as erythromycin, particularly in patients receiving anticoagulants, antiarrhythmics, statins, or psychotropic medications [91]. Excluding NTM infection before treatment initiation remains essential [1, 2].

Inhaled antibiotics may be considered for treating chronic *P. aeruginosa* infections, although the evidence supporting their effectiveness is inconclusive [1, 2]. In older adults, practical barriers to administration, such as impaired dexterity, cognitive impairment, treatment burden, and difficulty operating nebulizers or inhalers, may limit adherence and reduce the effectiveness of treatment [92]. Particular attention should also be paid to antibiotic-related toxicity. Aminoglycosides may increase the risk of nephrotoxicity and ototoxicity, especially in patients with chronic kidney disease or pre-existing hearing impairment [93]. Fluoroquinolones should be prescribed cautiously, as older age is associated with an increased risk of adverse effects, including tendinopathy, dysglycemia, central nervous system toxicity, and QT prolongation [94].

Emerging anti-inflammatory therapies, such as brensocatib, a dipeptidyl peptidase-1 inhibitor that targets neutrophil serine protease activation, have demonstrated reductions in exacerbation frequency (rate ratio 0.79; 95% CI 0.68–0.92) and airway inflammatory activity [95]. Brensocatib was recently licensed in the US and Europe but was not included in the 2025 European Respiratory Society guidelines because it was not approved in time for inclusion. Nevertheless, data regarding long-term safety, tolerability, and optimal dosing in older adults are limited, underscoring the need for dedicated studies in this population.

Polypharmacy is highly prevalent among older adults, often exceeding 15 medications [96]. Structured medication review is therefore essential to identify potentially inappropriate prescriptions, minimize drug–drug interactions, simplify treatment regimens, and reduce the overall treatment burden. Dose adjustments based on renal and hepatic function should be routinely considered [80]. While no studies have evaluated specific strategies for reducing medication use in older patients with bronchiectasis, evidence from broader geriatric populations indicates that reducing medication use is generally feasible and safe, with low rates of adverse withdrawal effects [97]. In clinical practice, clinicians should regularly reassess the balance between benefits and harms, prioritize treatments supported by clear evidence of efficacy, and integrate medication optimization into comprehensive geriatric assessments. However, no studies have evaluated the safety and efficacy of these strategies in older patients with bronchiectasis. Further research is therefore needed to establish bronchiectasis-specific strategies for reducing medication use and to optimize pharmacological management in this growing patient population.

Although there is little direct evidence regarding older adults with bronchiectasis, international guidelines recommend pulmonary rehabilitation [21], which has been shown to improve exercise capacity and health-related quality of life [98]. Due to the high prevalence of frailty, sarcopenia, and functional decline in older patients, comprehensive rehabilitation programs that incorporate aerobic and resistance training, balance training, education, and nutritional interventions [99, 100] may be particularly relevant for this population.

## 8 Prognosis and Outcomes

Older age is associated with increased mortality, reduced quality of life, and higher healthcare utilization. A study of 14,283 Korean National Health Insurance Service patients found that those with bronchiectasis over 60 years old had a significantly higher 10-year mortality rate than an age-matched cohort (2505 vs 2142 per 100,000 person-years;  $p < 0.001$ ). The greatest mortality risk was among the oldest patients (HR 1.17, 95% CI 1.10–1.25) [101].

European registry data showed that age, low BMI, prior hospitalizations, and reduced FEV<sub>1</sub> independently predicted 3-year mortality [15]. Similarly, US Bronchiectasis Research Registry data identified FEV<sub>1</sub> <50% of the predicted value, hospitalization within 2 years, and BMI <18.5 as key mortality predictors [1].

The higher comorbidity burden in older patients likely contributes to these adverse outcomes. Specific comorbidities significantly increase the risk of mortality relative to matched controls, including lung cancer (adjusted HR 1.85), pneumonia (HR 1.50), cardiovascular disease (HR 1.34), COPD (HR 1.24), and asthma (HR 1.20) [101]. Notably, patients without bronchiectasis-related comorbidities did not show a significant difference in mortality compared with the control group. This finding underscores the central role of multimorbidity in driving poor outcomes.

Older patients with bronchiectasis demonstrate substantially higher healthcare utilization compared with age-matched controls. A US Medicare study of 92,529 patients (mean age 76.7 years) found that patients with bronchiectasis had significantly more physician visits (15.4 vs 13.2), pulmonologist visits (1.9 vs 0.3), hospitalizations (0.4 vs 0.3), and higher total healthcare costs (\$10,224 vs \$6704) than matched controls [102]. Healthcare costs increased with exacerbation frequency—\$8506 for patients with no baseline exacerbations, \$10,365 for patients with one baseline exacerbation, and \$14,790 for patients with two or more baseline exacerbations [81]. In the EMBARC registry, 26.4% of patients were hospitalized at least once for an exacerbation in the preceding year, with 17.0% being hospitalized once, 5.7% twice, and 3.7% three or

more times [9]. There is substantial regional variation; patients in Central and Eastern Europe had significantly higher hospitalization rates (57.9% vs 26.4% overall), more severe disease (51.3% vs 35.1% with severe BSI), and increased exacerbation frequency (adjusted RR 1.12) and hospitalizations (adjusted RR 1.71) compared with patients in other regions [9]. *P. aeruginosa* infection increases management costs by an additional \$31,551 to \$56,499, primarily due to hospitalization [103].

## 9 Conclusion

Older adults with bronchiectasis are an important clinical population [15, 24]. They are characterized by age-related biological changes, such as immunosenescence and inflammation. They also have an increased burden of multimorbidity, frailty, and functional decline. However, it is uncertain whether these features define a true age-specific biological phenotype. Although the primary respiratory symptoms are similar to those observed in younger populations, diagnosis in older patients is often delayed due to symptom overlap with conditions such as COPD and heart failure. This delay in diagnosis contributes to worse outcomes, including higher mortality rates, lower quality of life, and greater utilization of healthcare resources [15, 18].

Importantly, disparities in outcomes in this population appear to be predominantly driven by frailty and multimorbidity rather than chronological age, bronchiectasis severity, or microbial burden alone [15]. Thus, management strategies should go beyond disease-specific interventions and incorporate a comprehensive, individualized framework integrating respiratory care and a systematic evaluation of comorbidities, frailty, and functional status [19, 24]. In this context, a treatable traits approach may be valuable because it emphasizes identifying and managing modifiable pulmonary and extrapulmonary factors contributing to symptoms, exacerbations, and functional decline [104]. Relevant age-related traits may include chronic airway infection, impaired mucus clearance, aspiration, GERD, malnutrition, sarcopenia, psychological comorbidities, and chronic rhinosinusitis.

Therefore, therapeutic decision making should balance efficacy with safety. This requires accounting for age-related pharmacokinetic and pharmacodynamic changes, comorbidity burden, and patient-centered priorities, such as preserving functional independence and quality of life. Interventions targeting frailty, such as pulmonary rehabilitation, structured exercise training, and nutritional optimization, may provide additional clinical benefits.

In summary, optimizing the management of bronchiectasis in older adults requires a multidisciplinary, individualized approach that integrates pharmacological

and nonpharmacological interventions while addressing multimorbidity, frailty, and functional impairment. This approach can help minimize treatment-related harm and maximize functional status, quality of life, and other patient-centered outcomes.

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## Declarations

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**Consent to participate** Not applicable.

**Consent for publication** Not applicable.

**Availability of data and materials** Not applicable.

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**Author contributions** MC and MGM supervised the project. All authors searched and screened all relevant literature. MC and MGM conducted the data extraction and collection. All authors performed the data analysis and review. MC and MGM drafted the manuscript. FP, RP, and VP critically revised the content. All authors read and approved the final version of the manuscript.

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